



In a scenario where we are using freely available public wireless networks all the time, security considerations are vital. Wireless attacks can be launched on the unwary and unprepared in several ways. An evil twin attack is one such serious threat.

The era of wireless connectivity offers us flexibility and mobility, but also comes with security issues. With wired connectivity, the user is more secure due to the physical connectivity. In wireless connectivity, however, the user is completely dependent on the AP or the SSID. In this article, I will discuss a very serious attack called the 'evil twin'.

Before proceeding further, you will need to get familiar with the terminology used in this article.

Service Set Identifier (SSID): An SSID is a 32-character (maximum) alphanumeric key identifying the name of the wireless local area network (WLAN).

AP (Access Point): This is used to connect the wireless device to the wireless network.

BSSID: This is the MAC address of the AP.

In an evil twin attack, the attacker makes a clone of a legitimate AP, and tries to route the user traffic from the cloned AP by means of any kind of DOS of the legitimate AP.

An evil twin attack can be launched in a cafe, hotel or any open wireless access area. Consider a scenario in which you connect to the familiar AP's SSID -- you know only its name. Next time, when you start your PC, it searches for the known AP. Evil twin attacks leverage preferred network lists, which enable automatic re-association with previously-used networks.

In this article, I will demonstrate how to make a clone of an AP, then its association with the client and finally, I will

discuss the defence techniques suggested by researchers.

Launching an evil twin attack

To launch the attack, I used the following:

- Kali Linux as the attacker
- Windows 7 as the victim
- A wireless AP

Kali Linux was installed in a VMware machine with a USB wireless card (Atheros-based), and it successfully recognised the card. I then followed the steps given below to launch the attack.

Step 1: To know the name of the wireless card, I typed the command *airmon-ng* as shown in Figure 1.

The wireless card's name is wlan0.

Step 2: The next command *airmon-ng start wlan0* set the wireless card in monitor mode 0, which means mon0 as shown in Figure 2.

Step 3: Then I ran the command *airodump-ng mon0*. Figure 3 shows the effect of the command. It also shows all the APs in the vicinity of the attacker. The red rectangle shows the target AP's SSID (*L4wisdom.com*), channel number (1) and BSSID (0C:D2:B5:01:0F:E6).

Step 4: Here, I set mon0 to Channel 1, using the following command (as shown in Figure 4):

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airodump-ng mon0 -c <channel> --bssid <mac address of AP>
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